

1 Solution — GIAM 3.5

Problem 6

1.1 Problem

A *vampire number* is a $2n$ -digit number v that factors as $v = xy$, where x and y are n -digit numbers and the digits of v are the union of the digits in x and y in some order. The numbers x and y are the “fangs” of v . To eliminate trivial cases, pairs of trailing zeros are disallowed.

- Show that there are no 2-digit vampire numbers.
- Show that there are seven 4-digit vampire numbers.

1.2 Answer

No 2-digit ones: $y = 10x/(x - 1)$ is never a digit.

Seven 4-digit ones: 1260, 1395, 1435, 1530, 1827, 2187, 6880.

1.3 Part One: No 2-Digit Vampire Numbers

Here $n = 1$, so the fangs x and y are single digits and v is a two-digit number whose digits are x and y .

Neither fang can be 0, since that would make $v = 0$. So $1 \leq x, y \leq 9$.

Theorem. *There is no 2-digit vampire number.*

Proof. Suppose $v = xy$ is one. Its two digits are x and y , so either $v = 10x + y$ or $v = 10y + x$. The product xy is symmetric in x and y , so these two possibilities differ only by which fang we call x ; **without loss of generality** assume

$$v = 10x + y = xy.$$

If $x = 1$ this reads $10 + y = y$, which is impossible. So assume $x \geq 2$ and rearrange:

$$10x = xy - y = y(x - 1), \quad \text{hence} \quad y = \frac{10x}{x - 1}.$$

Since y is an integer, $x - 1$ divides $10x$. But $\gcd(x - 1, x) = 1$, so $x - 1$ divides 10 , giving

$$x - 1 \in \{1, 2, 5, 10\}, \quad x \in \{2, 3, 6, 11\}.$$

As x is a digit, only $x \in \{2, 3, 6\}$ survive. Checking each:

$$x = 2 \Rightarrow y = \frac{20}{1} = 20, \quad x = 3 \Rightarrow y = \frac{30}{2} = 15, \quad x = 6 \Rightarrow y = \frac{60}{5} = 12.$$

None of these is a single digit, so every case fails and no such v exists. ■

Confirmed independently by testing all 81 products of two digits.

1.4 Part Two: Exactly Seven 4-Digit Vampire Numbers

Now $n = 2$, so $10 \leq x, y \leq 99$ and $1000 \leq v \leq 9999$. Taking $x \leq y$ to avoid counting each pair twice, there are $\binom{90}{2} + 90 = 4095$ candidate pairs — a finite check. Testing each for the digit condition and discarding pairs where **both** fangs end in 0 yields exactly seven values of v :

v	fangs	digit check
1260	21×60	$\{0, 1, 2, 6\} = \{2, 1\} \cup \{6, 0\}$
1395	15×93	$\{1, 3, 5, 9\} = \{1, 5\} \cup \{9, 3\}$
1435	35×41	$\{1, 3, 4, 5\} = \{3, 5\} \cup \{4, 1\}$
1530	30×51	$\{0, 1, 3, 5\} = \{3, 0\} \cup \{5, 1\}$
1827	21×87	$\{1, 2, 7, 8\} = \{2, 1\} \cup \{8, 7\}$
2187	27×81	$\{1, 2, 7, 8\} = \{2, 7\} \cup \{8, 1\}$
6880	80×86	$\{0, 6, 8, 8\} = \{8, 0\} \cup \{8, 6\}$

Each row can be checked by hand: multiply the fangs, then compare the digit multisets. For instance $21 \times 60 = 1260$, and the digits 2, 1, 6, 0 of the fangs rearrange to 1, 2, 6, 0. ■

1.5 The Picture

THE DEFINITION

A $2n$ -digit number v is a *vampire number* if $v = xy$ with x and y each n digits, and the digits of v are exactly the digits of x together with those of y , rearranged. Pairs of fangs both ending in 0 are disallowed.

PART ONE: NO 2-DIGIT VAMPIRE NUMBERS

Here x and y are single digits, so v has digits x and y in one of the two orders. Since the product xy is symmetric, the two orders differ only by renaming, and we may assume without loss of generality that

$$10x + y = xy \Rightarrow 10x = y(x - 1) \Rightarrow y = 10x / (x - 1)$$

Since $\gcd(x - 1, x) = 1$, the factor $x - 1$ must divide 10, so $x - 1 \in \{1, 2, 5, 10\}$ and $x \in \{2, 3, 6, 11\}$.

Only $x \leq 9$ is allowed, leaving three cases (and $x = 1$ separately, where $10 + y = y$ is absurd):

$$x = 2 \rightarrow y = 20$$

$$x = 3 \rightarrow y = 15$$

$$x = 6 \rightarrow y = 12$$

None of these y is a single digit, so no 2-digit vampire number exists. ■

Confirmed by checking all 81 products of two digits.

PART TWO: EXACTLY SEVEN 4-DIGIT VAMPIRE NUMBERS

from all 4095 pairs $10 \leq x \leq y \leq 99$

v	fangs $x \times y$	digits of v	digits of x and y
1260	21×60	0 1 2 6	0 1 2 6
1395	15×93	1 3 5 9	1 3 5 9
1435	35×41	1 3 4 5	1 3 4 5
1530	30×51	0 1 3 5	0 1 3 5
1827	21×87	1 2 7 8	1 2 7 8
2187	27×81	1 2 7 8	1 2 7 8
6880	80×86	0 6 8 8	0 6 8 8

THREE THINGS WORTH NOTICING

1. The trailing-zero rule changes nothing here. Fangs 10a and 10b give 100ab, whose digits are those of ab plus two zeros, so it would need ab to be a 2-digit vampire — and Part One says there are none.
2. A single trailing zero is fine: $1260 = 21 \times 60$ and $1530 = 30 \times 51$ both use one.
3. Each of the seven has exactly one fang pair. Ambiguity starts later: $125460 = 204 \times 615 = 246 \times 510$.

Figure 1.1: A diagram in four parts. The first states the definition: a $2n$ -digit number v is a vampire number if v equals x times y with x and y each n digits, and the digits of v are exactly the digits of x together with those of y , rearranged; pairs of fangs both ending in 0 are disallowed. The second part treats Part One, that there are no 2-digit vampire numbers. Here x and y are single digits, so v has digits x and y in one of the two orders; since the product xy is symmetric, the two orders differ only by renaming, and we may assume without loss of generality that $10x$ plus y equals xy , which gives $10x$ equals y times x minus 1, so y equals $10x$ over x minus 1. Since the greatest common divisor of x minus 1 and x is 1, the factor x minus 1 must divide 10, so x minus 1 lies in the set 1, 2, 5, 10 and x lies in 2, 3, 6, 11. Only x at most 9 is allowed, leaving three cases, with x equals 1 handled separately since 10 plus y equals y is absurd. The three cases give x equals 2 with y equals 20, x

equals 3 with y equals 15, and x equals 6 with y equals 12. None of these y is a single digit, so no 2-digit vampire number exists, confirmed by checking all 81 products of two digits. The third part is Part Two, exactly seven 4-digit vampire numbers, found from all pairs with x at most y between 10 and 99. A table lists each with its fangs, the digits of v , and the digits of x and y : 1260 equals 21 times 60; 1395 equals 15 times 93; 1435 equals 35 times 41; 1530 equals 30 times 51; 1827 equals 21 times 87; 2187 equals 27 times 81; and 6880 equals 80 times 86. In every row the two digit lists agree. The final band gives three things worth noticing. First, the trailing-zero rule changes nothing here, because fangs 10a and 10b give 100ab, whose digits are those of ab plus two zeros, so it would need ab to be a 2-digit vampire, and Part One says there are none. Second, a single trailing zero is fine, as in 1260 equals 21 times 60 and 1530 equals 30 times 51. Third, each of the seven has exactly one fang pair; ambiguity starts later, since 125460 equals 204 times 615 and also 246 times 510.

1.6 Remark — The WLOG Is Doing Real Work

Part One is a textbook use of the section’s second technique. The two-digit number v could be $10x + y$ or $10y + x$, and at first sight both must be examined. The WLOG collapses them into one because of a genuine symmetry:

the hypothesis $v = xy$ is symmetric in x and y ,

so swapping the labels turns any solution of $v = 10y + x$ into a solution of $v = 10x + y$ with the names exchanged. Since we are asking only whether a solution *exists*, the relabelled version is just as good.

It is worth stating the check one should always perform before writing “WLOG”: **is the situation genuinely unchanged by the relabelling?** Here it is, because the constraint set — “ x, y are digits” and “ $v = xy$ ” — is invariant under the swap. Had the problem said something asymmetric, such as $x < y$, the WLOG would have been illegitimate and both cases would need separate treatment.

1.7 Remark — Why $x - 1$ Divides 10

The step from $y = \frac{10x}{x-1}$ to “ $x - 1$ divides 10” deserves a second look, since it is where an infinite-looking search becomes four cases.

Because y is an integer, $(x - 1) \mid 10x$. Now x and $x - 1$ are consecutive, hence co-prime, so $x - 1$ shares no factor with x and must divide the remaining factor 10. Equivalently, write

$$\frac{10x}{x-1} = \frac{10(x-1) + 10}{x-1} = 10 + \frac{10}{x-1},$$

which is an integer exactly when $x - 1 \mid 10$. This second form also shows immediately that $y = 10 + \frac{10}{x-1} > 10$ for every $x \geq 2$ — so y can *never* be a digit, and the three cases are not even needed. The theorem could be stated in one line: for $x \geq 2$ the required y exceeds 10, and $x = 1$ is absurd.

Both routes are legitimate; the case version is more in the spirit of the section, and the inequality version shows why the cases were always going to fail.

1.8 Remark — The Trailing-Zero Rule Is Vacuous at This Size

The definition excludes fangs that both end in 0, and it is natural to ask what that rule is doing. At $n = 2$ it is doing nothing at all.

Suppose both fangs end in zero: $x = 10a$ and $y = 10b$ with a, b digits. Then

$$v = 100ab,$$

so the digits of v are the digits of ab followed by two zeros, while the digits of the fangs are $\{a, 0, b, 0\}$. Matching them requires ab to be a two-digit number with digits a and b — that is, a **2-digit vampire number**. Part One says there is none, so no such v exists and the rule excludes nothing. Running the search with and without the rule returns the same seven numbers, confirming this.

The rule earns its keep at larger sizes, where it prevents inflating any vampire number into a longer one by appending zeros to both fangs. Note also that a *single* trailing zero is perfectly acceptable, and two of the seven use one: $1260 = 21 \times 60$ and $1530 = 30 \times 51$.

1.9 Remark — What Happens Next

The seven 4-digit examples are the complete beginning of a sparse and irregular sequence. Extending the same search:

digits $2n$	vampire numbers	density
2	0	0
4	7	$7/9000 \approx 0.08\%$
6	148	$148/900000 \approx 0.016\%$

Both counts were computed directly. So they thin out sharply — unsurprising, since the digit condition is a strong coincidence and products of n -digit numbers spread over a range far larger than the supply of matching digit patterns.

Two further observations from the 6-digit data. All seven 4-digit vampires have a *unique* fang pair, but ambiguity appears at six digits: $125460 = 204 \times 615 = 246 \times 510$ is the only one below a million with two decompositions. And **1260**, the first entry in the table, is the smallest vampire number of any size — the example Clifford Pickover used when he introduced the notion in 1994.

There is no known formula or characterisation; the sequence is defined by a digit coincidence rather than by arithmetic structure, so exhaustive search remains the only way to produce it. That makes this problem a fair illustration of a case analysis whose cases simply have to be checked — with Part One showing what it looks like when the cases can be reduced to three by algebra first.